

physics

The Physics of Lunar X-ray Fluorescence Spectroscopy

The measurement of elemental abundances on the Moon via X-ray spectroscopy is a chain of physical processes that begins 150 million kilometers away at the Sun and ends in a detector orbiting the Moon. This document details each step of that physical journey.

1. The Source: Solar Coronal X-rays

The process is powered by the Sun. The Sun's outer atmosphere, the **corona**, is a superheated plasma at temperatures of millions of Kelvin. At these extreme temperatures, two primary mechanisms produce the X-rays that travel to the Moon:

- **Bremsstrahlung (Braking Radiation):** In the dense coronal plasma, high-velocity free electrons are constantly interacting with ions. As an electron is deflected by the electric field of an ion, it "brakes," losing kinetic energy. This lost energy is radiated away as a photon. The collection of countless such interactions produces a smooth, **continuous spectrum** of X-rays that forms the underlying shape of the solar spectrum.
- **Line Emission:** The corona is so hot that atoms (especially heavier ones like Iron) are highly ionized, having lost many of their electrons. The remaining bound electrons can be excited to even higher energy levels by collisions. When these electrons de-excite, they fall back to lower energy levels, emitting photons with very specific, discrete energies. These appear as sharp **emission lines** on top of the Bremsstrahlung continuum.

The XSM (Solar X-ray Monitor) instrument's job is to measure this complete incident spectrum, $F_{\text{sun}}(E)$, which is the sum of both the continuum and the line emission.

2. The Interaction: Photons Meet the Lunar Regolith

When a solar X-ray photon, γ_{sun} , arrives at the Moon, it interacts with the atoms in the surface layer of rock and dust (the regolith). Two competing processes can occur:

2.1 Photoelectric Absorption (The "Signal" Process)

If the incoming photon has an energy E greater than the binding energy of an inner-shell electron of a lunar atom (e.g., the K-shell binding energy, E_K), the photon can be completely absorbed by the atom. Its energy is transferred to the electron, which is then ejected from the atom.

This process leaves the atom in an unstable, ionized state with a vacancy in its innermost electron shell. This is the crucial first step for fluorescence.

2.2 Scattering (The "Noise" Process)

Alternatively, the photon can "bounce off" an electron in a lunar atom. This scattering is a major source of contamination in our final spectrum.

- **Rayleigh (or Thomson) Scattering:** The photon scatters elastically, meaning it changes direction but loses virtually no energy ($E_{\text{out}} \approx E_{\text{in}}$). This is the dominant scattering process for the lower-energy X-rays relevant to this analysis.
- **Compton Scattering:** The photon scatters inelastically, transferring a portion of its energy to the electron. The scattered photon therefore has a lower energy ($E_{\text{out}} < E_{\text{in}}$).

Both processes contribute to the bright, scattered continuum, $S(E)$, that forms the background in the CLASS spectrum. Its shape is a modified version of the Sun's spectrum, $F_{\text{sun}}(E)$.

3. The Response: X-ray Fluorescence (XRF)

Following the photoelectric absorption described in 2.1, the atom is in an unstable state and must return to a lower energy state.

3.1 Atomic Relaxation & Characteristic Emission

The vacancy in the inner K-shell is immediately filled by an electron from an outer shell (e.g., the L-shell or M-shell). Since the outer-shell electron is at a higher

energy level, it must release energy as it "falls" into the inner-shell vacancy. This energy is emitted as a new photon, γ_{XRF} .

The energy of this new photon is discrete and unique to the element. It is equal to the difference in the binding energies of the two electron shells involved:

$$E_{\text{XRF}} = E_{\text{initial_shell}} - E_{\text{final_shell}} \quad (\text{e.g., } E_{\text{L}} - E_{\text{K}})$$

This emitted photon is the **characteristic X-ray** or **fluorescence line**.

- **K-alpha ($K\alpha$) line:** Produced by an L-shell electron falling into the K-shell.
- **K-beta ($K\beta$) line:** Produced by an M-shell electron falling into the K-shell.

By measuring the precise energy of these $K\alpha$ lines (e.g., ~1.74 keV for Silicon, ~1.25 keV for Magnesium), we can unambiguously identify the elements on the lunar surface.

3.2 Fluorescence Yield

An important factor is that not every K-shell vacancy produces an XRF photon. The atom can also de-excite by emitting another electron (an "Auger electron"). The probability that a vacancy will result in an XRF photon is called the **Fluorescence Yield (ω)**. This value, which is unique to each element, is a crucial parameter in the full equations used to convert a measured line flux into an absolute elemental abundance.

4. The Measurement: How the Detector Records the Signal

The final step is the measurement of these photons by an orbiting detector like CLASS. A real detector is not a perfect instrument; its physical properties modify the signal it records.

4.1 Effective Area (ARF)

The **Ancillary Response File (ARF)** describes the instrument's **effective area**, $A(E)$. This is not just the physical size of the detector, but its overall efficiency at detecting a photon of a specific energy E . It's a combination of the geometric area, the quantum efficiency of the detector material (the probability that a photon of energy E will be absorbed and create a signal), and the transmission of any filters. It has units of cm^2 .

4.2 Energy Dispersion (RMF)

The **Response Matrix File (RMF)** describes the instrument's imperfect **energy resolution**. A beam of photons all with a perfect, single energy E will not be recorded in a single detector channel. Due to statistical processes in the detector, their energy is measured with some uncertainty, and they are recorded in a distribution of channels that is typically shaped like a Gaussian. The RMF, $R_i(E)$, is the probability matrix that describes how a true photon of energy E gets "blurred" or "redistributed" into the observed detector channel i . This physical effect is why we fit the elemental lines with Gaussian peaks, not infinitely sharp lines.